

1. Administrative & Study Identification

Full Study Title: A Phase 2, Multicenter, Randomized, Double-blind, Placebo-Controlled, Study to Evaluate the Efficacy and Safety of BMS-986256 in Participants with Active Systemic Lupus Erythematosus **Short Title/Acronym:** IM026-024 / Afimetoran in Active SLE **Protocol Number:** IM026-024 **NCT Number:** NCT04895696 **Sponsor Name:** Bristol-Myers Squibb (Bristol-Myers Squibb International Corporation) **Investigational Product:** Afimetoran (BMS-986256) **Phase of Development:** Phase 2 (Phase 2b) **Medical Monitor:** Head of the Global Submission Management - Clinical Trials (mg-gsm-ct@bms.com)

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the effectiveness (response rate per the SLE Responder Index (SRI) at Week 48), safety, and tolerability of Afimetoran in participants with active Systemic Lupus Erythematosus (SLE). **Secondary Objectives:** To evaluate SRI(4) response with corticosteroid (CS) reduction and maintenance to ≤ 7.5 mg per day, British Isles Lupus Assessment Group-based Composite Lupus Assessment response, reduction in Cutaneous Lupus Erythematosus Disease Area and Severity Index (CLASI) scores, reduction in counts of swollen and tender joints, and patient-reported outcomes.

2.2 Background & Rationale Activation of toll-like receptor (TLR) 7/8 by self-RNA induces the production of type I interferon and proinflammatory cytokines, contributing to the pathophysiology of SLE and related autoimmune diseases. Afimetoran is a novel, oral, potent, and selective inhibitor of TLR 7/8 designed to reduce the overactivity of the immune system seen in SLE. This study aims to determine whether TLR7/8 inhibition with Afimetoran leads to meaningful clinical improvement in SLE disease activity compared to placebo.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** 1:1:1:1 (Afimetoran Dose Level 1, DL2, DL3, or placebo)

3.2 Planned Enrollment & Duration Target N: Approximately 344 participants **Number of Sites:** Global multicenter study **Estimated Study Start:** May 2021 (based on registry history) **Estimated Primary Completion:** Ongoing

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Diagnosed ≥ 12 weeks before the screening visit and qualify as having SLE according to the SLE International Collaborating Clinics (SLICC) Classification Criteria at the screening visit. 2. Test positive, as determined by the central laboratory, for at least one of the following lupus-related autoantibodies at the time of screening: antinuclear antibody (ANA) $\geq 1:80$, anti-double-stranded deoxyribonucleic acid (dsDNA) antibody, or anti-Smith antibody. 3. Have a total Hybrid Systemic Lupus Erythematosus Disease Activity Index (SLEDAI) score ≥ 6 points and clinical Hybrid SLEDAI score ≥ 4 points with joint involvement and/or rash.

4.2 Exclusion Criteria 1. Active severe lupus nephritis (LN) as assessed by the investigator. 2. Active or unstable neuropsychiatric lupus manifestations defined by the Hybrid SLEDAI. 3. Diagnosis of Mixed Connective Tissue Disease for which the predominant diagnosis is not SLE, or Antiphospholipid Syndrome.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Oral Afimetoran (BMS-986256) at doses of 2.5 mg, 10 mg, or 30 mg, or matching placebo administered once daily. **Route of Administration:** Oral **Duration of Treatment:** 48 weeks (followed by an optional extension period).

5.2 Concomitant & Prohibited Therapy Permitted: Standard baseline medications for SLE, with a protocol-defined corticosteroid (CS) tapering therapy to reduce placebo responses. **Prohibited:** Other investigational systemic or biological therapies unapproved for concurrent use in this trial protocol.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Up to Day 0	Informed Consent, Medical History, SLICC/SLEDAI assessments, autoantibody laboratory testing
Baseline	Day 0	Randomization, First Dose
Treatment / Interim	Weeks 1 to 48	Up to 14 treatment visits over 48 weeks for physical exams, photography of affected skin, ECG, vital signs, and SLE disease assessments
End of Study	Week 48	Final SRI(4) primary endpoint assessment, safety evaluations, optional transition to the open-label extension period

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Response rate per the SLE Responder Index (SRI) at Week 48. **Measurement Method:** Composite clinical responder index calculated from SLEDAI, BILAG, and Physician's Global Assessment scores.

7.2 Secondary & Exploratory Endpoints * SRI(4) response with corticosteroid (CS) reduction and maintenance to ≤ 7.5 mg per day. * British Isles Lupus Assessment Group-based Composite Lupus Assessment response. * Reduction in Cutaneous Lupus Erythematosus Disease Area and Severity Index (CLASI) scores. * Reduction in counts of swollen and tender joints. * Patient-reported outcomes and overall safety/tolerability profile.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous evaluation during the 14 scheduled visits over 48 weeks via physical examinations, ECGs, and vital signs monitoring. **Serious Adverse Events (SAEs):** Standard 24-hour reporting to the Sponsor/Regulatory bodies. **Data Monitoring Committee (DMC):** Ongoing review by a panel of lupus trial experts to ensure the accuracy, safety, and consistency of patient data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy analysis; Safety Set for all participants receiving at least one dose of the investigational product. **Sample Size Justification:** Approximately 344 participants globally, adequately powered across 4 arms (1:1:1:1 ratio) to detect statistically significant differences in SRI(4) response rates at Week 48. **Handling of Missing Data:** Dealt with per the Statistical Analysis Plan (SAP), typically utilizing non-responder imputation (NRI) for primary composite indices like SRI(4).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits across the global sites. **Audit Trail:** eCRF locking, tracking of safety reports, and secondary continuous review by a dedicated panel of lupus trial experts.

11. Study Identifiers & Governance

Primary ID: IM026-024 **Registry Number:** NCT04895696 **Sponsor/Lead Organization:** Bristol-Myers Squibb **Study Title:** A Phase 2, Multicenter, Randomized, Double-blind, Placebo-Controlled, Study to Evaluate the Efficacy and Safety of BMS-986256 in Participants with Active Systemic Lupus Erythematosus **Official Regulatory Title:** A Study Evaluating the Efficacy and Safety of Afimotoran Compared With Placebo in Participants With Active Systemic Lupus Erythematosus (SLE)

12. Clinical Framework

Primary Condition: Systemic Lupus Erythematosus (SLE) **Diagnosis Threshold:** Total Hybrid SLEDAI score ≥ 6 points and clinical Hybrid SLEDAI score ≥ 4 points with joint involvement and/or rash. **Medical Methodology:** TLR 7/8 pathway inhibition via oral administration of Afimotoran. **Optimization Target:** Achievement of SRI(4) response criteria and subsequent corticosteroid tapering to ≤ 7.5 mg/day.

13. Study Procedures & Methodology

Management Pathway: Standard treatment visits (up to 14 over 48 weeks) evaluating disease scores (SLEDAI, BILAG, CLASI). **Education Protocol (HCP):** Site initiation visits and ongoing lupus trial expert reviews. **Education Protocol (Patient):** Informed consent detailing double-blind probability (75% Afimotoran / 25% Placebo) and required visit schedules. **Quality Audit Mechanism:** Dedicated expert panel reviews for accuracy and consistency of patient data and clinical assessments.

14. Observation & Follow-Up Schedule

Total Duration: 48 weeks (plus an extension phase) **Onsite Visit Cadence:** Up to 14 scheduled visits during the 48-week period **Remote Monitoring Frequency:** As outlined per the site-specific monitoring plan. **Checklist Review Interval:** Regular intervals coinciding with the 14 assessment visits.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	____	[DD-MMM-YYYY]				